

Sean Luka Girgis

Observability Platform Engineer | Telemetry, APM, Python Automation & Cloud Monitoring

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Professional Summary

Observability and performance engineer with deep enterprise APM, telemetry, dashboarding, alerting, RCA, and automation experience in large-scale banking and enterprise environments. At Citi, built Python, SQL, Pandas, and PySpark workflows that converted telemetry from 6,000+ infrastructure endpoints into dashboards, capacity forecasts, reliability reports, and operational insights. Strong hands-on foundation with Dynatrace AppMon/Synthetics, CA APM/Introscope, AppDynamics, BMC TrueSight/TSCO, Python/shell automation, production troubleshooting, cloud-oriented monitoring workflows, and modern observability tools including OpenTelemetry, Prometheus, Grafana, and Terraform/IaC ramp-up.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Supported enterprise observability, capacity, and reliability workflows across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and infrastructure telemetry from 6,000+ endpoints.
- Built Python, Pandas, PySpark, and SQL workflows that transformed telemetry into validated dashboards, KPI reports, forecasting datasets, operational insights, and reliability decision support.
- Analyzed utilization trends, latency symptoms, missing feeds, abnormal metric movement, and reporting discrepancies to identify bottleneck risk, reliability concerns, and capacity pressure.
- Created validation checks for telemetry freshness, row counts, missing data, unmatched records, abnormal deltas, and output consistency to improve confidence in monitoring and reporting data.
- Troubleshooted production issues by tracing source feeds, metric behavior, Python processing, SQL transformations, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Developed runbooks, metric definitions, dashboard notes, data-flow documentation, validation checkpoints, and support procedures to improve repeatable operational response.
- Partnered with infrastructure, application, analytics, engineering, and business teams to explain telemetry findings, coordinate fixes, and improve production reliability.
- Used AWS S3, Glue, Athena, Redshift, EC2/ECS concepts, Python, SQL, and monitoring data to support cloud-oriented observability, automation, and reporting patterns.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, improving visibility into user-facing reliability and performance.
- Correlated real-user monitoring, synthetic transaction data, and load-test behavior to identify bottlenecks, user-impacting symptoms, and platform improvement opportunities.
- Led the FAST project, mining real-user and synthetic monitoring telemetry to surface optimization recommendations adopted by engineering teams.
- Created dashboards and reporting views that converted raw monitoring signals into actionable performance, reliability, and stakeholder insights.

- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and performance reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Defined monitoring standards, dashboards, alert rules, SLA thresholds, validation checks, and operational reporting artifacts for application, middleware, infrastructure, and operations teams.
- Provided production troubleshooting guidance using CA APM telemetry, transaction visibility, dashboards, alerts, J2EE/WebLogic monitoring, and performance-analysis patterns.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable observability processes.
- Delivered onboarding support, training material, technical documentation, and knowledge sharing to improve adoption of monitoring and troubleshooting practices.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and production-readiness testing.
- Documented performance baselines, defects, risks, telemetry findings, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led validation and production-readiness support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OC4J/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, Pandas, PySpark, Perl, KornShell/ksh, Unix/Linux scripting

Cloud & Data: AWS S3, Glue, Athena, Redshift, EC2/ECS concepts, CloudWatch concepts, telemetry pipelines, data validation, reporting datasets, cloud monitoring concepts

AI/ML: Capacity forecasting, anomaly detection concepts, service-health reporting, Prophet/scikit-learn exposure, RAG/FastAPI project exposure

Tools & Infra: OpenTelemetry, Prometheus, Grafana, Terraform/IaC ramp-up, Docker, CI/CD concepts, Dynatrace AppMon, Gomez Synthetic Monitoring, CA APM / Introscope, AppDynamics, BMC TrueSight / TSCO, JMX Monitoring

Legacy: C, C++, Pro*C, Oracle RAC, PL/SQL, DB2, MQ Series, WebLogic/WebSphere, J2EE, performance testing, production troubleshooting

Flagship Projects

Enterprise APM Reporting Automation

Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry across large financial-services monitoring environments.

Technologies: CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation

FAST Performance Analytics Project

Mined Dynatrace and Gomez Synthetic Monitoring telemetry to identify optimization opportunities, improve transaction visibility, and communicate performance recommendations to engineering teams.

Technologies: Dynatrace AppMon, Gomez Synthetic Monitoring, Performance Analytics, Dashboards, APM

Production Telemetry Quality and RCA Workflow

Built validation and troubleshooting workflows for telemetry-backed reporting datasets, including freshness checks, missing feed diagnosis, abnormal trend detection, prior-run comparisons, and runbook documentation.

Technologies: Python, SQL, Pandas, BMC TrueSight, AppDynamics, RCA, Runbooks

ServiceCall AI RAG Demo

Built staged Python proof-of-concepts with FastAPI service layers, Docker-first smoke tests, Pydantic contracts, deterministic endpoints, and testable service behavior.

Technologies: Python, FastAPI, Docker, Pydantic, pytest, RAG